

2 Numbers are stored in a computer system using binary floating-point representation with:

- 12 bits for the mantissa
- 4 bits for the exponent
- two's complement form for both the mantissa and the exponent
- 2 bytes in total.

(a) Outline the effect of changing the number of bits used to store the mantissa to 10 bits, in this system.

.....

.....

.....

.....

.....

.....

..... [3]

(b) Describe **one** example of a situation that could cause an overflow to occur and the consequences of such an overflow.

.....

.....

.....

.....

.....

..... [3]

3 Describe how packet switching is used to pass messages across a network. Do **not** include checking for completeness and resending packets in your answer.

.....

.....

.....

.....

.....

.....

.....

..... [4]

